Exercise 5

On related time rates

Interactive test

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Choose the correct answer from those given :**

(1) The length of a rectangle is twice its width expands regularly preserving its shape and the ratio between its dimensions remains constant , the rate of change of its length is 6 cm. /sec. , then the rate of change of its width is ..... cm./sec.

(a) 12 (b) 3 (c) 6 (d) – 3

(2) The rate of increase of side length of an equilateral triangle is 2 cm. /sec. such that the triangle preserves its shape, then the rate of increase of its perimeter is ..... cm. /sec.

(a) 2 (b) 8 (c) 4 (d) 6

(3) A square of side length 5 cm. , the rate of increase of its side is 2 cm./sec., the length of its side after t seconds equals ..... cm.

(a) 2 t (b) 5 + 2 t (c) 2 t – 5 (d) $5 + 4\,t^2$

(4) If $y = x^2 - 3x$, then $\frac{dy}{dt} = \frac{dx}{dt}$ at $x =$ .....

(a) 1 (b) 2 (c) 3 (d) 4

(5) *(2nd session 2022)* A point moves on the curve : $y^2 + x^2 + 2x = 7$. If $\frac{dx}{dt} = 3$ at the point (1 , 2) which lies on the curve, then the rate of change of y-coordinate with respect to time at the same point = ...

(a) – 4 (b) – 3 (c) 3 (d) 4

(6) A point moves on the curve $y = 2x + 1$, then the ratio between the rate of change of the x-coordinate with respect to time for a point to the rate of change of the y-coordinate with respect to time equals .....

(a) 2 (b) – 2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

2 A point moves on a curve whose equation is $x^2 + y^2 - 4x + 8y - 6 = 0$. If the rate of change of its x-coordinate with respect to time at point (3 , 1) equals 4 units/sec. Find the rate of change of its y-coordinate with respect to time t. « $-\frac{4}{5}$ »

3 *(1st session 2011)* A point (x, y) moves along the curve : $x^2 + y^2 + 2x - 4y = 27$ Determine the position of this point at the instant when the rate of change of x-coordinate with respect to time equals to the rate of change of y-coordinate with respect to time.

« $(-5, 6), (3, -2)$ »

4 *(2nd session 2008)* A point (x, y) moves along the curve $x^2 + 4x - y = 3$, determine the position of the point at the instant when the velocity of its y-coordinate is twice the velocity of its x-coordinate.

« $(-1, -6)$ »

5 *(2nd session 2001)* A particle moves along the curve : $y = x^2 - \frac{1}{4}$, determine the position of this particle at the instant when the rate of change of its y-coordinate with respect to the time equals three times the rate of change of its x-coordinate with respect to the time.

« $(1.5, 2)$ »

6 **Choose the correct answer from those given :**

(1) A square lamina expands regularly , the rate of increasing in surface area of the lamina 75 cm^2/sec. , then the rate of increasing of side length = ... cm./sec. when the side length is 5 cm.

(a) 2.5 (b) 5 (c) 7.5 (d) 15

(2) A metal lamina in a square form , the rate of increasing its side length 0.2 cm./sec. then the rate of change of its surface area = ... cm^2/sec. when its diagonal length $8\sqrt{2}$ cm.

(a) $\frac{1}{25}$ (b) 3.2 (c) $\frac{2}{5}$ (d) 16

(3) A stone fell in a motionless lake , a circular wave was generated , the rate of increase of its radius length is 3 cm./sec. , then the rate of increase of its area after 4 seconds equals ..... cm^2/sec.

(a) 8π (b) 72π (c) 12π (d) 24π

(4) The rate of increase of area of a circular disc is 0.2 cm^2/sec. then the rate of increase of its radius length = ... cm./sec. when the radius length 7 cm. ($\pi = \frac{22}{7}$)

(a) $\frac{7}{22}$ (b) $\frac{22}{7}$ (c) $\frac{1}{220}$ (d) 3.5

(5) A metal disc , the rate of decreasing of its diameter by cooling is 0.5 cm./sec. , then the decreasing rate of its surface area = ... cm./sec. when its radius 14 cm. ($\pi = \frac{22}{7}$)

(a) 5.5 (b) 11 (c) 16 (d) 22

(6) A cubic water tank of edge length 4 m. containes some water , water is poured in it by rate $\frac{1}{2}$ m^3/min. then ,

First : The rate of increasing of water height in the tank = ... m./min.

(a) $\frac{1}{96}$ (b) $\frac{1}{32}$ (c) $\frac{1}{24}$ (d) $\frac{1}{48}$

Second : The rate of change of the area of the top surface of water = ... m^2/min.

(a) zero (b) 4 (c) $\frac{1}{4}$ (d) $\frac{1}{32}$

(7) A point moves on the curve : $y = x - \frac{x}{x^2+1}$, the rate of change of its x-coordinate with respect to time equals 9 at $x = \sqrt{2}$, then the rate of change of its y-coordinate with respect to time at the same point = ...

(a) 4 (b) 6 (c) 8 (d) 10

(8) A cube melt preserving its shape by rate 1 cm^3/sec , then the rate of change of its edge length when its volume 8 cm^3 is ..... cm./sec.

(a) $\frac{1}{12}$ (b) $\frac{1}{24}$ (c) $\frac{-1}{24}$ (d) $\frac{-1}{12}$

(9) A circle touches the sides of a square internally , then the rate of change of the radius length of the circle equals ..... the rate of change of the side length of the square at any instant given that the circle keeps touching the square internally.

(a) double (b) half (c) quarter (d) four times

(10) A square inscribed in a circle , then the rate of change of the radius of the circle equals ..... the rate of change of the square side at any point given that the vertices of the square stay on the circle.

(a) 1 (b) $\frac{\sqrt{2}}{2}$ (c) $\frac{1}{2}$ (d) $\sqrt{2}$

(11) The side length of a cube decreases by rate 3 cm./sec. , then the length of the diagonal of the cube is decreasing at a rate ..... cm./sec.

(a) $3\sqrt{2}$ (b) $3\sqrt{3}$ (c) 6 (d) 9

(12) If the side length of an equilateral triangle = a and it increases at a rate (k) , then the rate of increase of the triangle area equals .....

(a) $\frac{2}{\sqrt{3}}$ a k (b) $\sqrt{3}$ a k (c) $\frac{\sqrt{3}}{2}$ a k (d) $\frac{\sqrt{2}}{\sqrt{3}}$ a k

(13) A right circular cylinder expands preserving its shape , the rate of increasing of its radius length is 0.5 cm./sec. and its height (h) increase at a rate of 0.25 cm./sec. then the rate of change of its volume when r = 3 cm. , h = 5 cm. equals ..... cm^3/sec.

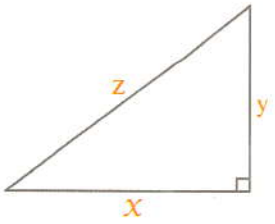
(a) $\frac{69}{4}\pi$ (b) $\frac{15}{4}\pi$ (c) $\frac{13}{2}\pi$ (d) $\frac{3}{4}\pi$

(14) In the opposite figure :

A right-angled triangle $\frac{dz}{dt} = 3$ cm./sec.

, $\frac{dx}{dt} = 3\frac{dy}{dt}$ then $\frac{dy}{dt} =$ ..... cm./sec.

at the instant when $x = 8$ cm. , $y = 6$ cm.



(a) 1 (b) 2 (c) 3 (d) 4

7 *(2nd session 2004)* A metallic lamina in the form of an equilateral triangle is expanding such that it preserves its geometrical shape. If the rate of increase of its side length is 0.1 cm./sec. , find its side length at the moment when the rate of increase of its area becomes $\sqrt{3}$ cm^2/sec. « 20 cm. »

8 The length of a rectangle is twice its width. It expands regularly such that it preserves its shape and the ratio between its dimensions. Find the area of the rectangle when its length increases at rate of 0.01 cm./sec. and its area increase at rate of 0.2 cm^2/sec. « 200 cm^2 »

9 A rectangular piece of metal , its length more than its width by 20 cm. , shrinking by cooling such that the length is always 20 cm. more than its width. the length decreases at a rate of 0.025 cm./sec. when its width 80 cm. Calculate the rate of change of its area at this moment. « – 4.5 cm^2/sec. »

10 *(2nd session 2003)* A is the area of the region bounded between two concentric circles with radii r_1 and r_2 , where $r_2 > r_1$. If r_1 is increasing with the rate 0.4 cm./sec. and r_2 is decreasing with the rate 0.2 cm./sec. , find the rate of change of A with respect to time t at the instant at which $r_1 = 8$ cm. and $r_2 = 14$ cm. « -12π cm^2/sec. »

11 *(2nd session 2009)* If stone fell in still water , then a circular wave is formed whose radius increases at a rate of 2.1 cm./sec. If the rate of increase of the surface area of the wave at the end of t seconds from the beginning equals 277.2 cm^2/sec. Find the value of t. ($\pi = \frac{22}{7}$) « 10 sec. »

12 *(2nd session 2010)* If the area of an isosceles right angled triangle equals (A) cm^2 and if $\frac{dA}{dt} = 32$ cm^2/sec. at the instant at which the length of each of the two equal legs of this triangle = 8 cm. , find (at this instant) the time rate of change of the perimeter (P) of the triangle. « $8 + 4\sqrt{2}$ cm./sec. »

13 A cube extends by heat so its edge length increases at a rate 0.02 cm./min. , and its surface area increases at a rate of 0.72 cm^2./min. , at a moment. Find the cube edge length at this moment and the rate of the increase in its volume at this time. « 3 cm. , 0.54 cm^3./min. »

14 A cube extends by heat so its edge length increases at rate 0.01 cm./min. , the rate of change of its volume at a moment is 75 cm^3./min. find.

(1) The edge length of the cube at this moment.

(2) The rate of change of its surface area at this moment. « 50 cm. , 6 cm^2./min. »

15 Two ships move from the same point on the same time , the first ship in direction of east with velocity 60 km./h. and the second in direction of south with velocity 80 km./h. Find the rate of change between the two ships after 2 hours from the beginning of motion

« 100 km./h. »

16 A rop passes over a small smooth pulley fixed at the top of a tower 12 m. high , a worker holds one of its ends at the base of the tower and the other ends fasten at a heavy car on the ground. If the worker can pull the rop at a rate 2 m./min. , calculate the speed of the car during its motion towards the tower when the car is at a distance 9 m. from the base of the tower. « $\frac{10}{3}$ m./min. »

17 A man stands at a distance 50 m. from a helicopter at the airport , the helicopter flies vertically up at a rate 65 m./sec. Find the rate of change of the distance between the man and helicopter when the plane 120 m. high from the ground surface. « 60 m./sec. »

18 A balloon rises up vertically with constant velocity 15 m./sec. , when the balloon at height 90 m. , a car passes on the ground just below the balloon and remains moving in a straight line with constant speed 25 m./sec. Find the rate of increasing of distance between the car and the balloon after 2 seconds from the moment that the car was below the balloon.

« $\frac{305}{13}$ m./sec. »

19 A balloon rises vertically from point A on the ground surface. An apparatus is placed to follow up the motion of the balloon at point B at the same horizontal plane of point A and distant 200 meters from it. At a moment , the apparatus observed the elevation angle of the balloon to find it $\frac{\pi}{4}$, and it increases at a rate of 0.12 rad /min. , find the rate of the balloon elevation at this moment. « 48 m. /min. »

20 A helicopter flies vertically at a fixed rate of 42 m./min. If a ground observer is 150 m. of the taking off site and observers the helicopter , find the rate of change of the elevation angle of the observer eyesight when the helicopter is 150 m. high from the ground. « $\frac{7}{50}$ rad. /min. »

21 *(1st session 2004)* A steel wire in the form of a right circular cylinder is heated such that its length increases at the rate of 0.005 cm./min. and its cross sectional diameter increases at the rate of 0.002 cm./min. Find in terms of π , the rate of change of its volume with respect to the time t when its length is of 40 cm. and its diameter is of 2 cm. long. « $0.085\ \pi$ cm^3/min. »

22 *(1st session 2006)* Water is poured into an empty cylindrical container of radius 10 cm. and height 60 cm. at the rate of $30\ \pi$ cm^3/sec. Find the rate at which the height of the water rises in the container. When does the container become full ? « 0.3 cm./sec. , 200 sec. »

23 A right circular cylinder- like oil tank whose diameter length of its base is 24 meters. If we want to empty the tank from the oil at a rate of 2 m^3/min. , what is the rate of the oil height in the tank? « $\frac{-1}{72\pi}$ m./min. »

24 The height of cylinder equals its base diameter , it expands such that the rate of increase of its surface area with respect to time equals $18\ \pi$ cm^2/sec. when the radius length of its base equals 6 cm. Calculate the rate of increase of its radius length. « $\frac{1}{4}$ cm./sec. »

25 A 2.6 meter Ladder is leaning against a vertical wall with its top and its base on horizontal ground. if the base of the ladder slides away from the wall at a rate of 4 m./min. when the ladder is 1 meter from the wall. Find the rate of the motion of the ladder top and the rate of change of the measure of the inclination angle of the ladder on the ground then. « $\frac{-5}{3}$ m./min. , $\frac{-5}{3}$ rad./min. »

26 A 6.5 metre ladder is leaning against a vertical wall with its upper end and its lower end on a horizontal ground , if the lower end slides away from the wall at a rate of 30 cm./ min. when it is at a distance 2.5 m. from the wall. Find the rate of descending of the ladder top , then find the height of the ladder top from the ground when the lower end and the upper end move with the same rate. « $\frac{1}{8}$ m./min. , $\frac{13\sqrt{2}}{4}$ m. »

27 A constant length ladder with upper end sliding on a vertical wall at a rate of k units/sec. , find the rate of sliding its base away from the wall when the ladder leans against the wall by an angle of θ where $\csc\theta = \frac{5}{4}$ « $\frac{3k}{4}$ unit./sec. »

28 A 4 meter ladder is leaning against a vertical wall with its top and its base on horizontal ground. If the base of the ladder slides away from the wall at a rate of 20 cm./sec. , calculate the rate of sliding the top of the ladder down when the ladder is inclined on the ground with an angle measured $\frac{\pi}{3}$ « $\frac{20\sqrt{3}}{3}$ cm./sec. »

29 Choose the correct answer from those given :

(1) The length of the radius of a circle increase at a rate $\frac{4}{\pi}$ cm./sec. , then the rate of increase of its perimeter at this moment is .....

(a) $\frac{4}{\pi}$ cm./sec. (b) $\frac{\pi}{4}$ cm./sec. (c) $\frac{1}{8}$ cm./sec. (d) 8 cm./sec.

(2) The rate of increase of radius length of a circle is 2 cm./min. and rate of increase of its area 20 π cm^2/min. , then the radius length at this moment equals ..... cm.

(a) $\frac{5}{2}$ (b) 5 (c) 10 (d) 20

(3) The length of a rectangle is twice its width , if the rate of increasing of its width is 3 cm./sec. , if the rectangle preserves the ratio between its dimensions then the rate of change of its diagonal length = ... cm./sec.

(a) 6 (b) $2\sqrt{5}$ (c) $3\sqrt{5}$ (d) $\sqrt{15}$

(4) If the height of an equilateral triangle increases at rate $\sqrt{3}$ cm./sec. , then the rate of change of its side length equals ..... cm./sec.

(a) 4 (b) 2 (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

(5) If x and y are functions of time (t) and $x + y$ = constant , then .....

(a) each of x and y increases at the same rate.

(b) each of x and y decreases at the same rate.

(c) one of them increases and the other decreases at the same rate.

(d) nothing of the previous.

(6) The rate of increase of surface area of a sphere is 6 cm^2/sec. at this moment its radius equals 30 cm. , then the rate of increase of its volume = ... cm^3/sec.

(a) 180 (b) 40 (c) 90 (d) 90 π

(7) An empty container , its volume 45 cm^3 , water is poured in it at a rate 5 cm^3/sec. , the container becomes full after ..... second.

(a) 9 (b) 135 (c) 45 (d) 5

(8) An isosceles triangle , the length of each of the two equal sides is 6 cm. and the measure of their included angle equals (x) , if the rate of change of (x) is $\left(\frac{\pi}{90}\right)^{rad}$ per minute , then the rate of change of its area at $x = \frac{\pi}{6}$ is .....

(a) $\frac{\sqrt{3}}{10}\pi$ (b) $\frac{\pi}{10}$ (c) $9\sqrt{3}$ (d) 9

(9) The relation between the vertical displacement (s) and time (t) sec is given by $s = 49\,t - 4.9\,t^2$, then the maximum displacement can the body reach after ..... sec.

(a) 9.8 (b) 10 (c) 5 (d) 50

(10) The slope of the tangent to the curve $y = f(x)$ at a point $= \frac{1}{2}$ and the x-coordinate of this point decreases at a rate 3 unit./sec. then the rate of change of its y-coordinate equals ..... unit./sec.

(a) $-\frac{1}{6}$ (b) $\frac{-3}{2}$ (c) $\frac{1}{6}$ (d) $\frac{3}{2}$

(11) When the rate of change of the area of a circle equals the rate of change of its diameter length numerically , then r = ...

(a) $\frac{2}{\pi}$ (b) $\frac{1}{\pi}$ (c) $\frac{\pi}{2}$ (d) π

(12) The side length of a cube increases at a rate 5 cm./sec. , then the volume of the cube increases at a rate ..... cm^3/sec. When its side length 10 cm.

(a) 1500 (b) 150 (c) 45 (d) 50

(13) A right circular cone expanding such that the height of the cone is always equal to its base diameter. If the rate of change of its base radius $= \frac{1}{\pi}$ cm./sec. then the rate of change of its volume = ... cm^3/sec. when its base radius length = 5 cm.

(a) 50π (b) $\frac{250}{3}\pi$ (c) 150 (d) 50

(14) If (A) is the area of a circle of radius length (r) , the radius length changes at constant rate , then .....

(a) A is constant. (b) $\frac{dA}{dt}$ is constant. (c) $\frac{dA}{dt} \propto r$ (d) $\frac{dA}{dt} \propto r^2$

(15) The number of sides of a regular polygon is m and its side length increases at constant rate a cm./sec. , then .....

(a) its perimeter increases at rate a cm./sec.

(b) its area increases at rate a cm^2/sec.

(c) its perimeter increases at rate m a cm./sec.

(d) its area increases at rate m a cm^2/sec.

(16) The number of sides of a regular polygon is m and its side length increases at constant rate a cm./sec. , then the angle at any vertex of the polygon .....

(a) increases at constant rate $(a)^{rad.}$/sec.

(b) increases at constant rate $(m\ a)^{rad.}$/sec.

(c) increases at non constant rate and can not be determined.

(d) remains constant.

(17) A man observes a plane flies horizontally at 3 km. high above him and with speed 480 km./h. , then the rate of change of the distance between the man and the plane after 30 sec. later = ...

(a) $\frac{320}{3}$ km./h. (b) 384 km./sec. (c) 384 m./sec. (d) $\frac{320}{3}$ m./sec.

(18) The radius length of a cylindrical tank is 25 cm. and its height 120 cm. , oil is poured in it at a rate $\frac{5000}{L+40}$ cm^3/sec. where L is the height of the oil at any moment then the rate of change of its height in the tank = ... cm./sec. when its half full.

(a) $\frac{4}{25}\pi$ (b) $\frac{1}{25\pi}$ (c) $\frac{2}{25\pi}$ (d) $\frac{8}{25\pi}$

(19) 10 meter ladder is leaning against a vertical wall and its lower end on a horizontal ground , if the lower end slides 2 m./min. then the rate of change of inclination angle with the horizontal at the moment the lower end at a distance 8 m. equals .....$^{rad.}$/min.

(a) 3 (b) – 3 (c) $\frac{1}{3}$ (d) $-\frac{1}{3}$

(20) An equilateral triangular lamina of side length 2 L , the side length increases by heat at a rate of $\frac{\sqrt{3}}{15}$ cm./sec. then the rate of increase of its area ..... cm^2/sec.

(a) $\frac{2}{5}$ L (b) $\frac{1}{5}$ L (c) $\frac{4}{5}$ L (d) $\frac{1}{10}$ L

(21) If y is positive and increases , then the value of y at which the rate of increase of y^3 equals 4 times the rate of increase of y is .....

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $2\sqrt{3}$ (d) $3\sqrt{2}$

(22) A circular segment , the radius length of its circle is 10 cm. and measure of its central angle (χ) changes in the rate 3 $^{rad.}$ per minute , then the rate of increasing in the area of the circular segment at $\chi = \frac{\pi}{3}$ is ..... cm^2/min.

(a) 125 (b) 75 (c) 150 (d) 300

30 The length of rectangle 12 cm. and its width 5 cm. the length decreases at a rate 1 cm./sec. while the width increases at a rate $\frac{1}{2}$ cm./sec. , find when does the shape become a square. Hence find the time the area stop increasing and find the area at this moment.

« $\frac{14}{3}$ sec. , 1 sec. , 60.5 cm^2 »

31 *(1st session 2014)* The width of a rectangle is increasing at a rate 1 cm./sec. while its length is decreasing so that the area of the rectangle remains constant and equals 24 cm^2 , find :

(1) The rate of change of the perimeter of the rectangle at the moment when the width equals 4 cm.

(2) The two dimensions of the rectangle at the moment when the perimeter stops changing.

« – 1 cm./sec. , $2\sqrt{6}$, $2\sqrt{6}$ cm »

32 *(1st session 2016)* A rectangle of length L and width W , has a constant area of 1200 cm^2 at a particular instant , the length is increasing at 6 cm./min. and the width is decreasing at 2 cm./min. Find the dimensions of the rectangle at this instant. Is the length of the diagonal of the rectangle increasing or decreasing ? At what rate ?

« 60 cm. , 20 cm. , $\frac{8\sqrt{10}}{5}$ cm./min. »

33 A metal triangular fine lamina expands regularly such that its base length equals three times its height the area increases at a rate 0.27 cm^2/sec. , find the rate of change of its height when its height 9 cm.

« 0.01 cm./sec. »

34 *(1st session 2012)* The length of the two sides of the right angle in a right-angled triangle at an instant were 15 cm. , 20 cm. the smaller side increases at a rate of 1 cm./minute and the larger side decreases at a rate of $\frac{1}{2}$ cm./minute.

(1) The rate of change of the surface area of the triangle at any instant (t) , then find its value when t = 5 minutes.

(2) The surface area of the triangle when its rate of change vanishes.

« 3.75 cm^2/min. , 189.0625 cm^2 »

35 *(2nd session 2013)* If the lengths of the sides of a right-angled triangle are changing while the perimeter remains constant at 40 cm. and the rate of change of the length of the hypotenuse is 7 cm./min. when the side lengths were 8 cm. , 15 cm. and 17 cm. , find the rate of change of the length of each of the other two sides at this instant.

« 25 cm./min. , –32 cm./min. »

36 An isosceles triangle whose base length is $20\sqrt{3}$ cm. If the length of the two equal legs decreases at a rate of 3 cm./h. , find the rate of decreasing the triangle surface area at the moment which the length of the two equal legs is equal to the base length. « 60 cm^2/h. »

37 Two sides in a triangle , the length of both increases at a rate of 0.1 cm./sec. and the measure of their included angle increases at a rate of $\frac{1}{5}$ rad./sec. at which rate does the area of the triangle change at the moment the length of each side of the triangle is 10 cm. ?

« 5.866 cm^2/sec. »

38 ABC is a right-angled triangle at C and its area is constant and equals 24 cm^2 , if the rate of change of b equals 1 cm./sec. find the rate of change for each of a and m (∠ A) at the moment in which b equals 8 cm.

« $\frac{-3}{4}$ cm./sec. , – 0.12 rad./sec. »

39 A metal fine lamina in rectangular form , its length is $\frac{4}{5}$ of its diagonal , shrinking by cooling preserving its geometrical shape and same ratio between its dimensions , at a certain moment , the length of its diagonal decreases at a rate 2.5 cm./min. , at the same moment its surface area decreases at a rate 60 cm^2/min. Find the surface area of the lamina at this moment.

« 300 cm^2 »

40 A boy holds a thread of a kite at 30 high , the wind pulls the kite horizontally at a rate 25 m./min. find the speed the boy should leave the thread when the thread length = 50 m.

« 20 m./min. »

41 A metal cuboid expands , its length 2 cm. more than its width and its height is 3 times its width by heating such that it preserves the ratio among its dimensions , the volume increases at a rate 0.6 cm^3/min. while its width increases at a rate 0.01 cm./min. , find the dimensions of the cuboid.

« 2 , 4 , 6 cm. »

42 A cuboid with square base , its base side increases at a rate 1 cm./min. and its height decreases at a rate 2 cm./min. Find the rate of increasing of its volume when its base side length 6 cm. and its height 24 cm. then find after how many minutes this increase will vanish.

« 6 minutes »

43 A metal regular quad. pyramid whose height equals its base side length. Its volume increases at a rate of 1 cm^3/sec. , if the rate of the increase of both the pyramid's height and its base side length equals 0.01 cm./sec. find its base side length.

« 10 cm. »

44 If the rate of increase of surface area of a circle is constant , prove that the rate of increasing of its circumference inversely proportion with its radius length.

45 A spherical balloon is filled with gas but the gas leaks at a rate of X cm^3/sec. , prove that the rate of the decrease of the balloon area at the moment which the radius length r cm. equals $\frac{2X}{r}$ cm^2/sec.

46 A point moves on the curve : $y = 4X - 2X^3$, its y- coordinate is decreasing at a rate of 5 units/sec. Find the rate of change of the slope of the curve at $X = 2$ « – 6 units./sec. »

47 A point starts to move from origin in the positive direction of X-axis with velocity 5 cm./sec. find the rate of change of the distance between this point and the point (9 , 8) after 3 seconds from the beginning of motion. « 3 cm./sec. »

48 A point moves on the curve of the function $f(X) = \sqrt{X^2 + 5}$ its X-coordinate increases at a rate of $3\sqrt{10}$ cm./sec. , find the rate of change of the distance between this point and the point (1 , 0) at $X = 2$ « 9 cm./sec. »

49 If point A (X , y) moves on the curve of the function $y = X^3 + X$ where $\frac{dy}{dt} = 2$ unit./sec. find the rate of change of the area of triangle AOB where (O) is the origin point and , B (0 , 6) at the moment at which X-coordinate of the moving point equals 3 « $\frac{3}{14}$ square unit / sec. »

50 A regular hexagonal – like lamina shrinks by cold. if the rate of change of its side length is 0.1 cm./sec. find the rate of change in the area of the lamina when its side length is 10 cm. « $-3\sqrt{3}$ cm^2/sec. »

51 A regular octagon whose side length is 10 cm. and it increases at a rate of 0.2 cm./sec. find the rate of the increase of its area. « 19.3 cm^2/sec. »

52 ***(2nd session 2014)*** A head lamp is placed at a height of 680 cm. from the surface of the earth. A man of a height 170 cm. walks with velocity 2 m./sec. in a straight line towards the base of the lamp. **Find :**

(1) The rate of change of length of the man's shadow.

(2) The velocity of the man's end of shadow « $\frac{-2}{3}$ m./sec. , $\frac{-8}{3}$ m./sec. »

53 *(2nd session 2005)* A man of height 180 cm. is walking with a velocity of magnitude 150 cm./sec. towards a lamp of height 480 cm. over the ground. **Find :**

(1) The rate of change of the length of the shadow of the man.

(2) The rate of change of the distance between the head of the man and the lamp at the instant at which the man is 300 cm. apart from the base of the lamp.

« -90 cm./sec. , $-75\sqrt{2}$ cm./sec. »

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54 A 180 cm. man moves far from the base of a 3 meter lamppost at a rate of 1.2 m./s. find the rate of change of the length of the man's shadow. If the straight line passing through the highest point of the man's head and the top of the lamp inclines on the ground with an angle of measure θ^{rad} when the man is far from the base of the lamppost for a distance x meter , prove that $x = \frac{6}{5}\cot\theta$, then find the rate of change of θ when the man is 3.6 m. from the base of the lamppost. « 1.8 m./sec. , $\left(\frac{-1}{10}\right)^{rad}$/sec. »

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55 A 170 cm. man ascends up an inclined plane with uniform velocity 6 m./min. , the plane makes angle whose tangent is $\frac{7}{24}$ and its length 25 m. , a lamp is fixed at height $11\frac{1}{4}$ m above the horizontal plane passes by the base of the inclined plane and vertically above the top of the inclined plane. Find the rate of decreasing of the man's shadow and also the rate of approaching of the end of the man's shadow to the top of the inclined plane.

« 4 m./min. , 10 m./min. »

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56 A 5 meter rod is fixed by a hinge to the ground at its base. If its top rises up by a winch at a rate of 1 m./min. , find the rate of decreasing the projector length of the rod on the ground when the height of the top is 3 meters. « $\frac{3}{4}$ m./min. »

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57 In a 100 meter race , the racer runs in a straight line toward the end line. If an end line camera is distant 5 meters and perpendicular to the racing pathway and at the same horizontal plane of the racers , find the rate of change of the angle by which the camera rotates to capture the racer when he was distant 5 meters from the end line and the rate of approaching the end point is 10 m./sec. « (-1) rad/sec. »

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58 *(2nd session 2012)* An iron sphere of diameter 12 cm. is covered with a layer of ice of uniform density and thickness. If the ice is melting at a rate of magnitude 20 cm^3/min. at the instant at which the ice thickness is decreasing at rate of $\frac{1}{20\pi}$ cm./min. , where π is the approximate ratio , and the ice keeps its spherical shape during melting , find at this instant each of the following :

(1) The thickness of ice layer.

(2) The rate of change of the outer surface area of the ice layer. « 4 cm. , 4 cm^2/min. »

59 A hollow sphere whose internal radius and external radius are r_1 , r_2 respectively. the inner radius increases at a rate of 1 cm./sec. and its volume remains constant at the moment $r_1 = 3$ cm. , $r_2 = 9$ cm. Find at this moment

(1) The rate of change of its external radius.

(2) The rate of change of its external surface area.

(3) The rate of change of its thickness. « $\frac{1}{9}$ cm./sec. , 8π cm.^{2}/sec. , $\frac{-8}{9}$ cm./sec. »

60 If the gas leaks from a spherical balloon at a rate of 20 cm.^{3}/sec. , find the rate of change of the balloon radius length at the moment in which the radius length is 10 cm. and find the rate of change of the balloon external surface area at the same moment.

« $\frac{-1}{20\pi}$ cm./sec. , -4 cm.^{2} / sec. »

61 **Choose the correct answer from those given :**

(1) If χ is the radian measure of an angle , $\chi \in]0 , \frac{\pi}{2}[$, then the rate of increasing of tangent of an angle equals 8 times the rate of increasing of sine the same angle at $\chi =$ .....

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{5}$ (d) $\frac{\pi}{6}$

(2) A spherical balloon whose radius r , its volume v , it is filled with gas but the gas leaks at constant rate , then .....

(a) $\frac{dr}{dt} > 0 , \frac{dv}{dt} > 0$ (b) $\frac{dr}{dt} > 0 , \frac{dv}{dt} < 0$

(c) $\frac{dr}{dt} < 0 , \frac{dv}{dt} > 0$ (d) $\frac{dr}{dt} < 0 , \frac{dv}{dt} < 0$

(3) A spherical balloon , its volume 100π cm.^{3} , full with gass and due to leacking of gass the volume of the balloon decreases at a rate of 8π cm.^{3}/min. Preserving its spherical shape Find :

First : The rate of change of its radius length when its radius = 4 cm. is ..... cm./min.

(a) $\frac{-1}{8}$ (b) $-\frac{1}{4}$ (c) $-\frac{1}{2}$ (d) -2

Second : The rate of change of its radius length after 8 minutes. from the beginning of leaking = ... cm./min.

(a) $\frac{-3}{8}$ (b) $\frac{-2}{9}$ (c) $\frac{-1}{9}$ (d) $\frac{-2}{3}$

(4) If the length of the base of a triangle (b) increases at a rate 3 cm./min. , and its corresponding height (h) decreases at a rate 3 cm./min. , then the area of the triangle (A) .....

(a) always increases. (b) remains constant.

(c) decreases only when $b < h$ (d) decreases only when $b > h$

(5) In the opposite figure :

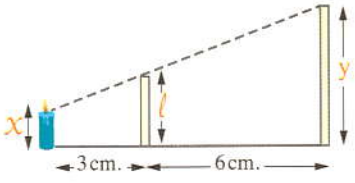


A lamp post of length 5.1 m. constructed on the top of an inclined road. A man 1.7 m. tall walks on the inclined road with a speed 2 m./sec. then the rate of increasing of the man shadow on the road = ... m./sec.

(a) 1 (b) 1.5

(c) 2 (d) 6

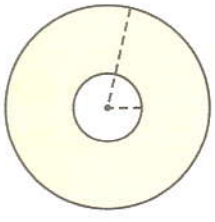
(6) *(Trial 2021)* In the opposite figure :



A dark barrier of height ℓ cm. is placed at a distance 3 cm. from a burning candle and at a distance 6 cm. from a vertical wall , if the height of the candle is decreasing by the rate 3 cm. / hr. , then the rate of changing of the shadow length (y) on the wall equals ..... cm./hr.

(a) 3 (b) – 3 (c) 6 (d) – 6

(7) In the opposite figure :

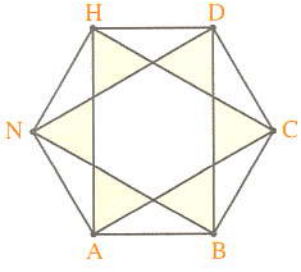


Two concentric circles , their radii are 6 cm. , 18 cm. If the radius of the smaller increases at a rate 2 cm./sec. and the radius of the greater increases at a rate 1 cm./sec. then the area between the two circles during the time interval]0 , 12[.....

(a) continuously increases. (b) continuously decreases.

(c) increases then decreases. (d) decreases then increases.

(8) *(1st session 2021)* ABCDHN is a regular hexagon , the length of its side increases at rate $\sqrt{3}$ cm./sec. , when the side length is 4 cm. , then the rate of change of the shaded region with respect to time = ... cm^2/sec.



(a) 12 (b) 8

(c) $12\sqrt{3}$ (d) $8\sqrt{3}$

62 *(1st session 2008)* The volume V of a sphere changes regularly such that it preserves its spherical shape. At any instant t sec. , the length of the radius of the sphere is r cm. its surface area is S cm^2 , prove that.

$\frac{dv}{dt} \times \frac{dr}{dt} = \frac{1}{16\pi} \times \left(\frac{ds}{dt}\right)^2$, then calculate the value of

$\frac{dr}{dt}$ when $\frac{dv}{dt} = \pi$ cm^3/sec. , $\frac{ds}{dt} = 2\pi$ cm^2/sec. ,

and find the surface area at that instant. « $\frac{1}{4}$ cm./sec. , 4π cm^2 »

63 A known mass of gas with a constant temperature whose volume decreases at a constant rate of 2 cm^3/sec. If the pressure is inversely proportional to the volume and the pressure equals 1000 gm.wt./cm^2 when the volume is 250 cm^3 , find the rate of change of the pressure with respect to time when the volume of gas is 100 cm^3 « 50 gm.wt. / cm^2/sec. »

64 In a closed electric circuit , if v is the potenital difference (volt) , I is the current intensity (Ampere), R is the resistance (ohm). If the potential difference increases at a rate of 1 volt/sec. and the current intensity decreases at a rate of $\frac{1}{2}$ Ampere/sec. , find the rate of the resistance at the moment which v = 12 volt and I = 2 Amperes. « 2 ohm/sec. »

65 From the origin point (O) , in the perpendicular coordinates plane a particle A moves in the direction 30° north of the east with velocity of magnitude 4 meter/min. after one minute another particle B moves from the same origin point (O) along the straight line $\overleftrightarrow{OB}$ whose its equation is $x + \sqrt{3}\, y = 0$
with velocity of magnitude 6 meter/min. in the direction which makes the angle $\angle$ BOA acute. Find the rate of change of the distance between the two particles A , B two minutes after the particle B started motion. «5 m./min. »

66 At 8 o'clock morning , a ship at a distance 60 km. east of a certain port approaching to it with velocity 10 km./h. At 9 o'clock morning another ship leaves the port towards south with velocity 30 km./h. Find the rate of change of the distance between the two ships at 10 o'clock morning and determine if they getting closer or away from each other? « 10 km./h. »

67 An airplane take off from cairo airport from point A. making angle of measure 60° with the horizontal to cover 15 km. in this direction then it flies horizontally away from the airport with velocity 750 km./h.
Find the rate of change of the distance between the plane and the point A after 2 minutes from the moment it flies horizontally. « $\frac{4875}{7}$ km./h. »

68 A ship sails from A with uniform velocity 18 km./h. towards north after two hours it sails in direction makes 30° north of east , find the rate of change of the distance between the ship and the point A after 4 hours from the beginning of sailing. « $9\sqrt{3}$ km./h. »

69 A ship sails from a port at 9 o'clock morning towards the west with velocity 20 km./h. and one hour later another ship sails from the same port with velocity 40 km./h. in direction 60° north of the west. Find the rate of diverge between the two ships at 11 o'clock morning. « 30 km./h. »

70 A man observes a boat moving in a straight line parallel to the river bank and 10 meters distant from it when the line connecting the man and the boat made an angle of measure 30° the rate of change of this angle equals $\frac{1}{2}$ rad./sec. Find the speed of the boat at this moment. « 20 m./sec. »

71 *(2nd session 2006)* $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are two intersecting roads enclosing an angle of measure 90°. A car traveled on the road $\overrightarrow{AB}$ starting from A with velocity 40 km./h. 10 minutes later another car traveled on the road $\overrightarrow{AC}$ starting from A with velocity 60 km./h. find the rate of change of the distance between the two cars 20 minutes after the second car started motion. « $50\sqrt{2}$ km./h. »

72 *(1st session 2007)* $\overrightarrow{AC}$ and $\overrightarrow{BC}$ are two perpendicular roads where AC = BC = 1000 m. two men started to walk , the first from A towards C with a uniform velocity of 50 m./min. , while the other from B towards C with a uniform velocity of 100 m./min. Prove that the distance between the two men after t minutes from the moment they started both to walk is given by the relation : $s^2 = (1000 - 50\ t)^2 + (1000 - 100\ t)^2$ **, then find :**

(1) The rate of change of the distance between the two men with respect to the time when t = 4

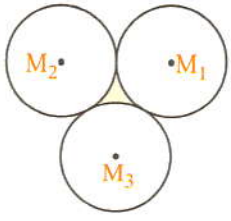
(2) When will the two men start to move away from each other ? « – 100 m./min. , 12 min. »

73 $\overrightarrow{OX}$, $\overrightarrow{OY}$ are two rays including angle of measure 150°. A is a point moves on $\overrightarrow{OX}$ with constant velocity 4 cm./sec. B is another point moves on $\overrightarrow{OY}$ such that the surface area of Δ OAB is constant and equals 30 cm^2. Find the magnitude and direction of B when OA = 8 cm. « 7.5 cm./sec. in direction $\overrightarrow{BO}$ »

74 50 metre rope passes over a pulley which is 24 m. high , one of its ends tied to a heavy mass and the other end tied to a car moves on the ground with velocity 18 m./sec. Find the rate of change of height of the mass at the moment when the car at a distance 32 m. from the projection of the pulley.

« 14.4 m./sec. »

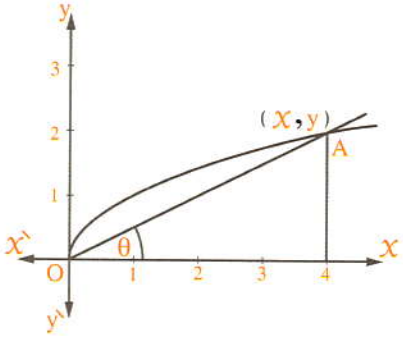
75 **In the opposite figure :**



3 identical circles touch each other two by two, the radius length of each is r , expands regularly such that they preserve their shape.If the radius length extends at rate 1 cm./min.
Find the rate of change in surface area of the shaded region at the moment when the radius length of each 10 cm.

« $\left(20\sqrt{3} - 10\,\pi\right)$ cm./min. »

76 **In the opposite figure :**



A particle (X , y) is moving along the curve of the function $y = \sqrt{X}$
When $X = 4$,
the y-component of the position of the particle is increasing at rate 1 length unit/sec.

(1) What is the rate of change of the X-component at this moment ?

(2) What is the rate of change of the distance from the origin to the particle at the same moment ?

(3) What is the rate of change of the angle of inclination θ at the same moment ?

« 4 length unit/sec. , $\frac{9\sqrt{5}}{5}$ length unit/sec. , $\left(\frac{-1}{5}\right)$ rad./sec. »

77 Two points A and B start to move together from origin (O) such that the point A moves on the positive X-axis with velocity 2 length unit/sec. and B moves in the first quadrant on the curve of the function $f : f(X) = X^2$ such that the length of $\overline{OB}$ = the length of $\overline{AB}$
Find the rate of change of area of triangle ABO after 5 seconds from the beginning of motion.

« 75 length unit/sec. »

78 The point A moves on the unit circle in the first quadrant and its X-coordinate moves with velocity 3 units/sec. Find the rate of change of the X-coordinate of intersection point between the tangent to the circle at A with the X-axis when the X-coordinate of the intersection point of the tangent with X-axis equals 2 units

« – 12 units/sec. »

79 If the point A moves in the positive direction of X-axis starting from the origin (O) with velocity $\frac{2}{3}$ length unit/min. and B (0 , 2) , C (0 , 4) Find the rate of change of the angle ($\angle$ BAC) when A reaches to (2 , 0) « $\frac{1}{30}$ rad/min. »

80 A 5-meter water pipe with two ends A and B is leaning with its end A on a horizontal ground and with one of its points D against a 3-meter vertical wall. If end A slides away from the wall at a rate of $\frac{5}{4}$ meters/min. , find the rate of sliding end B when the pipe reached the edge of the wall. « $\frac{3}{5}$ m./min. »

81 A right circular cylinder-like container of interior height 9 cm. and the interior radius length of its base is 6 cm. A 16-cm. metal rod is placed in the container. If the rate of sliding the rod away from the edge of the cylinder is 2 cm./s. find the rate of sliding the rod on the cylinder base when the rod reaches the end of its base. « $\frac{5}{2}$ cm./sec. »

82 Lighting lamp fixed at the top of a 16 m. tower if a ball is fallen from the same height from a window 15 m. distant from the lamp. Find the speed of the shadow of the ball on the ground when the ball at 6 m. high from the ground. « – 33.6 m./sec. »

83 A 15 m. lamppost , A ball is projected vertically up with velocity 5 m./sec. 12 m. away from the base of the lamppost , find the rate of diverge of the shadow on the ground from the base of the lamppost at the middle of the first second. « 0.09555 m./sec. »

Higher skills

84 **Choose the correct answer from those given :**

(1) The number of sides of non regular polygon = m and each side length increases at constant rate a cm./sec. , then .....

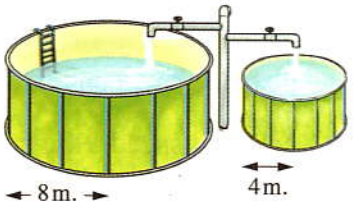
(a) its perimeter increases at rate a cm./sec.

(b) its perimeter increases at rate m a cm./sec.

(c) its perimeter increases but not at a constant rate.

(d) its area increases at rate a cm^2/sec.

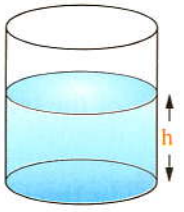
(2) Two cylindrical tanks of water , the radius length of the smaller is 4 m. and the radius length of the bigger one is 8 m. they are being filled simultaneously at the same rate , the rate of increases of water level in the smaller tank is $\frac{1}{2}$ m./min. , then the rate of increase of water level in the bigger tank = ... m./min.

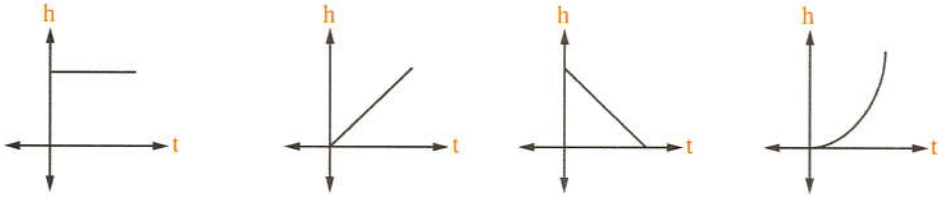


(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$

(3) In the opposite figure :

Water is poured into a right cylindrical tank at a constant rate , which of the following figures represents the relation between the water level (h) in the tank and the time (t) ?





(a) (b) (c) (d)

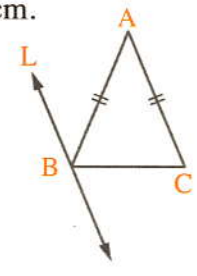
(4) A hemi - sphere tank of water , radius 2 m. , water is poured into it , if the rate of change of the height of water is $\frac{1}{4}$ m./min. , then the rate of change of the area of the water surface in the tank after 2 min. from the beginning of the pouring water is ..... m^2./min.



(a) $\frac{1}{4}\pi$ (b) $\frac{1}{2}\pi$ (c) $\frac{3}{4}\pi$ (d) $\frac{2}{3}\pi$

(5) In the opposite figure :

ABC is an isosceles triangle in which AB = AC = 13 cm. , BC = 10 cm. If the straight line L moves from point B parallel to $\overleftrightarrow{AC}$ in direction of $\overrightarrow{BC}$ to intersects $\overline{AB}$, $\overline{BC}$ at D , E respectively where the rate of change in BE = $\frac{1}{10}$ cm./sec. , then the rate of change of the area of Δ DEB at BE = 1 cm. equals ..... cm^2/sec.



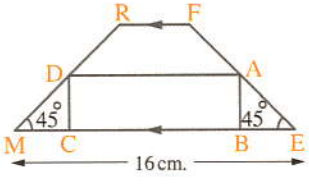
(a) 0.48 (b) 0.12 (c) 0.96 (d) 0.24

85 It is known that the light intensity at a point is proportion inversely to the square of the distance between that point and the light source. A boat moves towards a lighthouse with constant speed , the height of the lighthouse is 100 m. , prove that the rate of increasing of light intensity at the boat when it is 100 m. from the lighthouse base equals $\frac{25}{8}$ as much as when the boat at 200 m. from the lighthouse base.

86 ABC is an isosceles triangle , the length of its base BC = 6 cm. , and its altitude equals 8 cm. , $\overline{LM}$ is parallel to $\overline{BC}$ and cuts $\overline{AB}$ at L , $\overline{AC}$ at M. It is given that the height of the trapezium LMCB decreases at a rate of 0.5 cm./sec. find the rate of decreasing of the area of the trapezium LMCB when $\overline{LM}$ bisects each of the two sides $\overline{AB}$, $\overline{AC}$. « $-1.5\ cm^2/sec.$ »

87 **In the given figure :**

FEMR is an isosceles trapezium each side inclines 45° to the greater base , ABCD is a rectangle , A is moving from F to E and D from R to M , if the rate of change of length of AB = $\frac{1}{4}$ cm./sec. Find the rate of change of area of the rectangle ABCD when the length of AB = 3 cm. , and calculate the area of the rectangle when its area stop changing.



« $-1\ cm^2/sec.$, $32\ cm^2$ »

88 A right circular conical funnel , its height is 35 cm. , and the radius length of its base is 7 cm. fixed such that its axis is vertical while its apex is downward , oil is poured in it at a rate 5.4 cm^3/sec. while its leaking from a hole in the other end at a rate 3.2 cm^3/sec. find the rate of increasing of the height of oil when the oil height is 25 cm. $\left(\pi = \frac{22}{7}\right)$

«0.028 cm./sec. »

89 A right circular cone , its apex is placed downward the radius of its base is 80 cm. and its height is 160 cm. water is leaking out of it and collected in a cylindrical container , its base radius is 50 cm. Find the height of the water in the cone at the moment the rate of decreasing of water height in the cone equals the rate of increasing of water height in the cylinder.

« 100 cm. »

Problems related with solid geometry

90 A cuboid of dimensions 3 , 4 and 12 cm. If the rate of the increase of its first dimension is 2 cm./sec. and the second dimension is 1 cm./sec. but the rate of the decrease of its third dimension is 3 cm./sec.

Find : (1) 📖 The volume of the cuboid at any moment (t).

(2) 📖 The rate of change of its volume of the end of two second.

(3) The rate of change of the cuboid diagonal length at the same point.

« $-12\ cm^3/sec$, $\frac{2}{11}$ cm./sec. »

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91 A man across a bridge , 12 m. high above water surface , the speed of the man is 3 m./min. , the man observed a boat moving perpendicular to the bridge with a constant speed 6 m./min. exactly under the man. Find the rate of diverge between the man and the boat after 6 minutes from the moment that they are on the same vertical line.

« $\frac{45}{7}$ m./min. »

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92 A plane flies horizontally 2000 m high towards south with velocity 150 m./sec. , it passes over a car moves with velocity 75 m./sec. towards east. Find the rate of diverge of the plane from the car after 40 seconds.

« $160\ \frac{5}{7}$ m./sec. »